# Information Retrieval

## Lab 3: ***Inverted Index***

2020-2021

# Inverted Index

An inverted index is an index data structure storing a mapping from content.

It is a hash-map like data structure that directs you from a word to a document or a web page.

**<u>An inverted index contains two parts:</u>**

*index of terms (term index):* stores a distinct list of terms found in the collection

a *posting list:* a list of documents that contain the term.

- The size of the posting list indicates the number of documents that contain the term.
- A linked list is used to store the entries in a posting list

Word

Posting List

**Inverted index**

| ID | Term | Document |
|---|---|---|
| 1 | best | 2 |
| 2 | blue | 1, 3 |
| 3 | bright | 1, 3 |
| 4 | butterfly | 1 |
| 5 | breeze | 1 |
| 6 | forget | 2 |
| 7 | great | 2 |
| 8 | hangs | 1 |
| 9 | need | 3 |
| 10 | retire | 2 |
| 11 | search | 3 |
| 12 | sky | 2, 3 |
| 13 | wind | 2 |

# Inverted Index

Document ID (posting list)

Word

**Example:** Consider these two Documents:

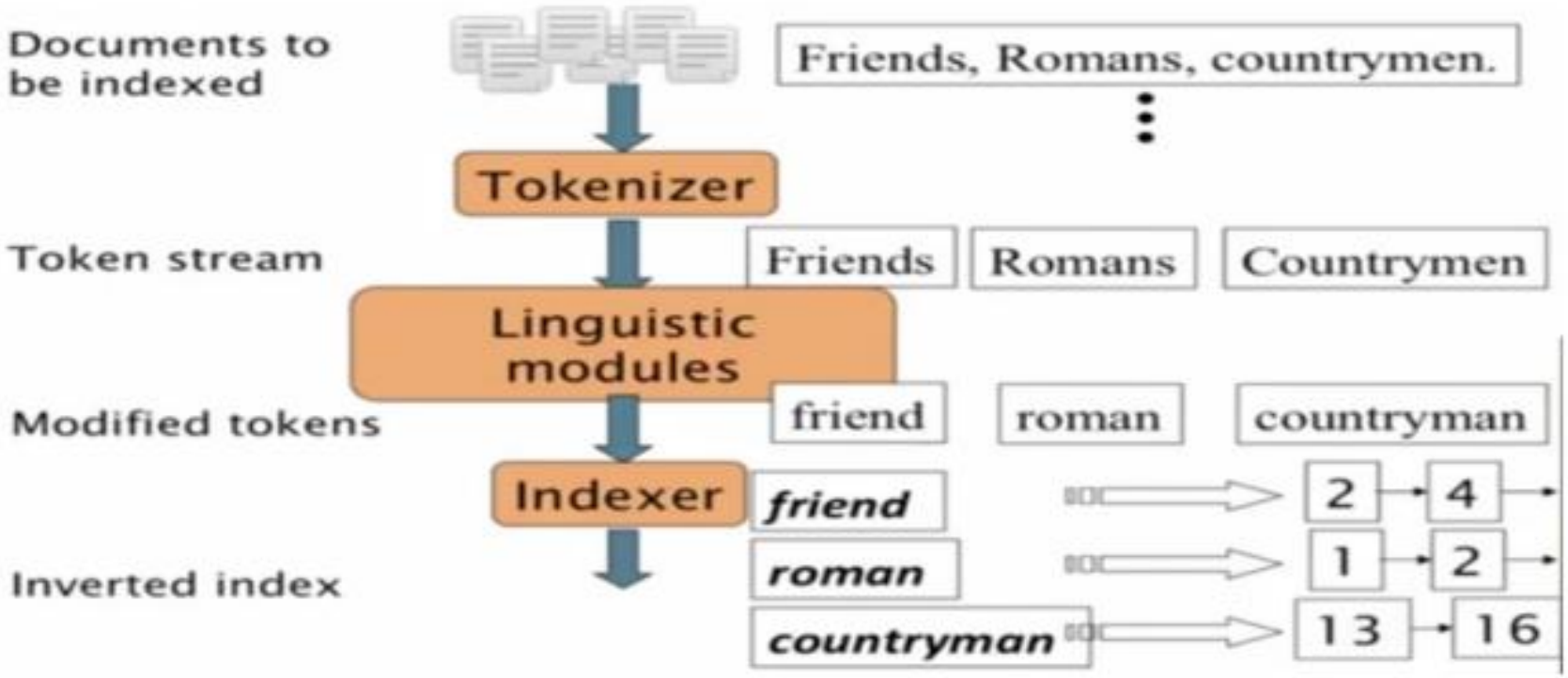
D1: The GDP increased 2 percent this quarter.

D2: The spring economic slowdown continued to spring downwards this quarter

**An inverted index for these two documents**

2 → [D1]
continued → [D2]
downwards → [D2]
economic → [D2]
GDP → [D1]
increased → [D1]
percent → [D1]
quarter → [D1] → [D2]
slowdown → [D2]
spring → [D2]
the → [D1] → [D2]
this → [D1] → [D2]
to → [D2]

# Index Construction



# Initial stages to build an inverted index:

**Fetch the Document:**

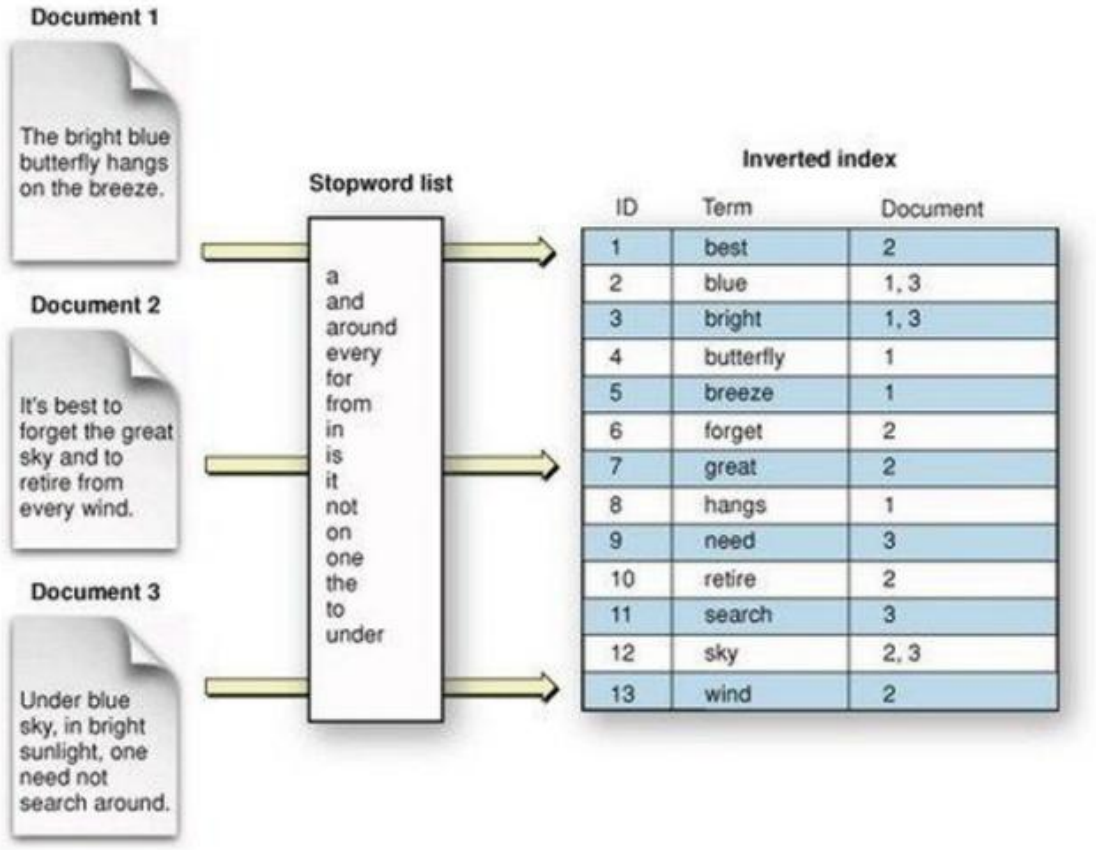
- Removing of Stop Words: Stop words are most occuring and useless words in document like “I”, “the”, “we”, “is”, “an”.

**Stemming of Root Word.**

- We may wish different forms of a root to match.
- **Ex: Cat and Cats**

**Record Document IDs**

- Add additional information like frequency of word, location of word etc.
- Repeat for all documents and sort the words.



# Document List

- Include the ***documentList*** variable in ***InvertedIndex*** *object.*
- The ***documentList*** is simply an **ArrayList** of Document objects.
- Its purpose is to map a document identifier to a Document object.
- The Document object contains information unique to each document, for example, theauthor, title, date published.
- Storing the list of document objects with the inverted index makes sense because the only time it is updated is when we update the inverted index.

# Example

**Document Collection**

| DocId | Document |
|---|---|
| 1 | Norway borders Sweden. Norway is to the west of Sweden. |
| 2 | Magnus Carlsen is a chess player from Norway. He is the world chess champion. |

| Term | DocId |
|---|---|
| norway | 1 |
| borders | 1 |
| sweden | 1 |
| norway | 1 |
| is | 1 |
| top | 1 |
| the | 1 |
| west | 1 |
| of | 1 |
| sweden | 1 |
| magnus | 2 |
| carlsen | 2 |
| is | 2 |
| a | 2 |
| chess | 2 |
| player | 2 |
| from | 2 |
| norway | 2 |
| he | 2 |
| is | 2 |
| the | 2 |
| world | 2 |
| chess | 2 |
| champion | 2 |

| Term | DocId |
|---|---|
| a | 2 |
| borders | 1 |
| carlsen | 2 |
| champion | 2 |
| chess | 2 |
| chess | 2 |
| from | 2 |
| he | 2 |
| is | 1 |
| is | 2 |
| is | 2 |
| magnus | 2 |
| norway | 1 |
| norway | 1 |
| norway | 2 |
| of | 1 |
| player | 2 |
| sweden | 1 |
| sweden | 1 |
| the | 1 |
| the | 2 |
| top | 1 |
| west | 1 |
| world | 2 |

**Inverted Index**

| Term | Doc Freq. | Postings Lists |
|---|---|---|
| a | 1 | [2] |
| borders | 1 | [1] |
| carlsen | 1 | [2] |
| champion | 1 | [2] |
| chess | 2 | [2] |
| from | 1 | [2] |
| he | 1 | [2] |
| is | 2 | [1,2] |
| magnus | 1 | [2] |
| norway | 2 | [1,2] |
| of | 1 | [1] |
| player | 1 | [2] |
| sweden | 1 | [1] |
| the | 2 | [1,2] |
| top | 1 | [2] |
| west | 1 | [1] |
| world | 1 | [2] |

**Boolean Queries**

norway OR champion

| | | | |
|---|---|---|---|
| | | v | |
| norway | 2 | 1 | 2 |
| champion | 1 | 2 | |
| | | ^ | |

norway AND champion

| | | | |
|---|---|---|---|
| | | v | |
| norway | 2 | 1 | 2 |
| champion | 1 | 2 | |
| | | ^ | |